

A Continuous Topological Interpretation of Quantum Circuits and Entanglement

Krishanu Karun

March 2026

Abstract

We introduce a continuous topological interpretation of quantum circuits that extends the usual discrete gate-based model. In this view, a circuit is not only a finite sequence of unitary operations, but a continuous landscape embedded in a larger parameter space. The standard unitary description appears as a constrained submanifold of this space. Within this framework, entanglement naturally shows up as an enlargement of the accessible parameter manifold, which helps explain why entangled circuits are significantly more expressive than collections of unentangled qubits.

1 Introduction

The standard description of a quantum circuit is fundamentally *discrete*. A circuit is specified by a finite list of quantum gates and measurements acting on a register of qubits, and each gate is a unitary operator on a Hilbert space. In variational and algorithmic settings, one usually optimizes expectation values of observables with respect to the parameters appearing in this discrete sequence of gates.

In this short note, we sketch a different way to look at the same objects. Instead of treating a circuit as a list of isolated unitaries, we regard it as a continuous landscape in an extended parameter space, with the usual unitary manifold sitting inside that space as the physically realizable slice. Intermediate values in this larger space do not have to be unitary; they can be thought of as points that are subsequently projected back onto the nearest unitary operation before implementation on hardware. We then extend this picture to entanglement and use it to argue, in geometric terms, why entanglement becomes essential for highly expressive quantum circuits.

2 Discrete object interpretation

Let $\mathcal{H}_N \cong \mathbb{C}^{2^N}$ be the Hilbert space of an N -qubit system. In the usual discrete interpretation, a parameterized single-qubit circuit can be written as

$$U(\boldsymbol{\theta}, \boldsymbol{\tau}) = \prod_{j=1}^m G_j(\theta_j, \tau_j), \quad (1)$$

where θ_j is a physical gate parameter (for example, a rotation angle) and τ_j is a temporal or ordering parameter associated with the j -th gate. For an n -qubit register, each gate also carries information about which qubits it acts on, so we write

$$U(\boldsymbol{\theta}, \boldsymbol{\tau}) = \prod_{j=1}^m G_{j, S_j}(\theta_j, \tau_j), \quad (2)$$

with $S_j \subseteq \{1, \dots, n\}$ denoting the subset of qubits addressed by G_j .

Given an initial state $|\psi_0\rangle$ and an observable o , the output state is

$$|\psi_{\text{out}}\rangle = U(\boldsymbol{\theta}, \boldsymbol{\tau}) |\psi_0\rangle, \quad (3)$$

and the corresponding objective (such as an energy, fidelity, or probability) is the expectation value

$$\langle O \rangle = \langle \psi_0 | U^\dagger(\boldsymbol{\theta}, \boldsymbol{\tau}) o U(\boldsymbol{\theta}, \boldsymbol{\tau}) | \psi_0 \rangle. \quad (4)$$

In this discrete picture, the whole structure of the circuit is encoded in a finite collection of unitary gates and their parameters, which live on the unitary manifold associated with $\mathcal{H}_N$.

3 Continuous topological interpretation

We now move from this discrete description to a continuous, topological one. Instead of viewing quantum states only as static vectors in $\mathcal{H}_N$, we treat them as continuous mappings on a parameter space, closer in spirit to functions over a high-dimensional domain. In this setting, we define a continuous circuit

$$U_{\text{cont}}(\boldsymbol{\lambda}) = \prod_{j=1}^m G_j(\boldsymbol{\lambda}_j), \quad (5)$$

where $\boldsymbol{\lambda} = (\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_m)$ collects all parameters of the circuit. Each block $\boldsymbol{\lambda}_j$ can itself be a vector, or in the limiting case an element of an infinite-dimensional space, reflecting the idea that very high-dimensional vectors can be regarded as functions.

The crucial point is that this extended parameter space is larger than the usual unitary manifold. The gates $G_j(\boldsymbol{\lambda}_j)$ are allowed to take values that are not directly unitary or immediately realizable in hardware. Conceptually, before each physical operation, one performs a projection or decomposition that maps these values back onto the closest point of the unitary manifold. In this way, the circuit can explore directions that would be invisible if one stayed strictly inside the discrete set of unitary gates.

In this continuous picture, it is natural to think of not only adjusting individual gate parameters, but shaping the entire landscape that the circuit traces out. This is reminiscent of continuous and topological optimization in classical fields (such as structural or architectural design), where one works inside a large design space and then enforces physical constraints afterwards. The same philosophy can be carried, at least at the conceptual level, into the design and analysis of quantum circuits.

4 Entanglement and parameter-space enlargement

The continuous viewpoint becomes more intuitive when we apply it to entanglement. A helpful picture is to imagine the full Hilbert space $\mathcal{H}_N$ as a large sheet of paper. Two unentangled subsystems, with Hilbert spaces $\mathcal{H}_n$ and $\mathcal{H}_m$, correspond to two separate regions (or circles) on this sheet. The total parameter space they can explore, without entanglement, has a dimension that is roughly the sum of the dimensions of their individual parameter spaces.

To make this more precise, consider splitting an n -qubit system into subsets $S_a \subseteq \{1, \dots, n\}$. For each subset we introduce local parameters

$$\boldsymbol{\lambda}_a^{\text{local}} = \{\theta_{a,j} ; j \in S_a\} \in \mathbb{R}^{d_a^{\text{local}}}, \quad (6)$$

and define local unitary blocks by

$$U_{S_a}^{\text{local}}(\boldsymbol{\lambda}_a^{\text{local}}) = \bigotimes_{j \in S_a} R_{\sigma_j}^{(j)}(\theta_{a,j}), \quad \sigma_j \in \{x, y, z\}. \quad (7)$$

We then consider states of the form

$$|\phi'_a(\boldsymbol{\lambda}_a^{\text{local}})\rangle = U_{S_a}^{\text{local}}(\boldsymbol{\lambda}_a^{\text{local}})|\phi_a(\boldsymbol{\lambda}_a^{\text{local}})\rangle, \quad (8)$$

with $\langle\phi_a|\phi_a\rangle = 1$, and build a global unentangled state

$$|\psi_{\text{unt}}\rangle = \bigotimes_{a=1}^n |\phi'_a(\boldsymbol{\lambda}_a^{\text{local}})\rangle. \quad (9)$$

The set of all such unentangled states forms a submanifold

$$\mathcal{M}_{\text{unt}} = \left\{ |\psi_{\text{unt}}\rangle ; \boldsymbol{\lambda}_a^{\text{local}} \in \mathbb{R}^{d_a^{\text{local}}}, \forall a \right\}, \quad (10)$$

with total parameter-space dimension

$$\sum_a d_a^{\text{local}}. \quad (11)$$

To add entanglement, we enlarge each parameter block by introducing entangling parameters

$$\boldsymbol{\lambda}_a^{\text{ent}} = \{\alpha_{a,k}\}_{k=1}^{d_a^{\text{ent}}} \in \mathbb{R}^{d_a^{\text{ent}}}, \quad (12)$$

and define entangling unitaries

$$U_{S_a}^{\text{ent}}(\boldsymbol{\lambda}_a^{\text{ent}}) = \exp(-i\alpha_a H_{S_a}), \quad \alpha_a \in \boldsymbol{\lambda}_a^{\text{ent}}, \quad (13)$$

for suitable multi-qubit Hamiltonians H_{S_a} . The combined block is

$$U_{S_a}(\boldsymbol{\lambda}_a) = U_{S_a}^{\text{local}}(\boldsymbol{\lambda}_a^{\text{local}}) U_{S_a}^{\text{ent}}(\boldsymbol{\lambda}_a^{\text{ent}}), \quad (14)$$

with

$$\boldsymbol{\lambda}_a = \{\boldsymbol{\lambda}_a^{\text{local}}, \boldsymbol{\lambda}_a^{\text{ent}}\} \in \mathbb{R}^{d_a}, \quad d_a = d_a^{\text{local}} + d_a^{\text{ent}}. \quad (15)$$

The full circuit is then

$$U_{\text{full}}(\boldsymbol{\lambda}) = \prod_{a=1}^m U_{S_a}(\boldsymbol{\lambda}_a), \quad (16)$$

and the resulting state

$$|\psi_{\text{full}}(\boldsymbol{\lambda})\rangle = U_{\text{full}}(\boldsymbol{\lambda}) |0\rangle^{\otimes n} \quad (17)$$

lies on an entangled submanifold

$$\mathcal{M}_{\text{ent}} = \left\{ |\psi_{\text{full}}\rangle ; \boldsymbol{\lambda} \in \Lambda \subseteq \mathbb{R}^D \right\}, \quad (18)$$

where

$$\Lambda = \prod_{a=1}^m (\mathbb{R}^{d_a^{\text{local}}} \times \mathbb{R}^{d_a^{\text{ent}}}) \cong \mathbb{R}^D, \quad D = \sum_a d_a, \quad (19)$$

and, by construction, there exist subsets S_a with $|S_a| > 1$, so genuine multi-qubit entangling operations are present.

Whenever at least one d_a^{ent} is nonzero, the parameter space underlying $\mathcal{M}_{\text{ent}}$ is strictly higher-dimensional than that of $\mathcal{M}_{\text{unt}}$. Returning to the paper analogy, the unentangled manifolds look like two separate circles on the sheet, while the entangled manifold behaves like a larger region that covers both circles and extends beyond them. In the continuous topological picture, entanglement is therefore not just a useful resource, but the mechanism that enlarges the landscape in which the circuit can move.

5 Outlook

The continuous topological interpretation outlined here treats quantum circuits as objects living in extended parameter landscapes, with the usual discrete gate model recovered as the unitary slice of that landscape. In this setting, entanglement appears as a natural way to increase the dimensionality of the accessible manifold, which offers a geometric explanation for the central role of entangled states in expressive quantum algorithms.

This perspective opens several possible directions. One can imagine optimization schemes that explicitly exploit the continuous structure of the landscape before projecting back onto hardware-friendly circuits, or borrow tools from topological and structural optimization in classical engineering. It may also provide a useful language for thinking about noise and imperfections, viewed as deformations in the extended parameter space that we try to steer back towards the desired unitary manifold.